

Question: 1

In Cisco Packet Tracer, create a simple two-switch network with three VLANs (e.g., VLAN 10 for 'Foodies', VLAN 20 for 'Gamers', VLAN 30 for 'Shoppers'), assign three PCs to each VLAN, and verify that PCs in the same VLAN can ping each other but not PCs in other VLANs.

Answer: 2

Cisco Packet Tracer Configuration Steps

- **Step 1: Set Up the Topology:** Drag and drop **two 2960 Switches** (Switch1, Switch2) and **nine PCs** into the workspace. Connect three PCs to Switch1 and six PCs to Switch2 using straight-through cables. Connect the two switches together via Gigabit interfaces (G0/1 to G0/1) using a cross-over cable.
- **Step 2: Assign IP Addresses:** Set a unique static IP address for all 9 PCs inside the same subnet domain (192.168.1.0/24) so they can theoretically communicate before isolating them with VLAN boundaries:
 - **VLAN 10 PCs:** 192.168.1.11, 192.168.1.12, 192.168.1.13
 - **VLAN 20 PCs:** 192.168.1.21, 192.168.1.22, 192.168.1.23
 - **VLAN 30 PCs:** 192.168.1.31, 192.168.1.32, 192.168.1.33
- **Step 3: Define VLANs on Both Switches:** Click on each switch, open the **CLI tab**, press Enter, and configure the database names exactly as requested:

text

```
Switch# configure terminal
```

```
Switch(config)# vlan 10
```

```
Switch(config-vlan)# name Foodies
```

```
Switch(config-vlan)# vlan 20
```

```
Switch(config-vlan)# name Gamers
```

```
Switch(config-vlan)# vlan 30
```

```
Switch(config-vlan)# name Shoppers
```

```
Switch(config-vlan)# exit
```

- **Step 4: Map Switch Ports to VLANs:** Assign the physical host port blocks into their matching structural network paths:

text

```
Switch(config)# interface range fa0/2-4
```

```
Switch(config-if-range)# switchport mode access  
Switch(config-if-range)# switchport access vlan 10
```

```
Switch(config)# interface range fa0/5-7  
Switch(config-if-range)# switchport mode access  
Switch(config-if-range)# switchport access vlan 20
```

```
Switch(config)# interface range fa0/8-10  
Switch(config-if-range)# switchport mode access  
Switch(config-if-range)# switchport access vlan 30  
Switch(config-if-range)# exit
```

- **Step 5: Enable the Trunk Link:** Configure the interface linking the two switches together to allow all VLAN traffic to pass across the link:

text

```
Switch(config)# interface g0/1  
Switch(config-e1000)# switchport mode trunk  
Switch(config-e1000)# exit
```

Verification Process

To verify the network segregation, open the **Command Prompt** on a PC and test the connectivity using ping commands:

- **Same VLAN Test:** On a VLAN 10 PC (192.168.1.11), run ping 192.168.1.12. The ping will succeed with a "**Reply from...**" message because they share the same broadcast domain.
- **Cross VLAN Test:** On the same PC, run ping 192.168.1.21 (a VLAN 20 PC). The ping will fail with a "**Request timed out**" message. This proves that the switches are isolating the traffic correctly, and devices cannot communicate across different VLANs without a router.

Question: 2

Configure trunk ports between the two switches in your Packet Tracer network so that all VLANs can communicate across both switches.

Hint: Use the 'switchport mode trunk' command and allow all VLANs on the trunk link.

Answer: 2

Steps to Configure Switch Trunking in Cisco Packet Tracer

To ensure that traffic from all three custom VLANs (Foodies, Gamers, and Shoppers) can flow between **Switch1** and **Switch2**, follow these precise command steps on the connecting port interface:

- **Step 1: Open the First Switch CLI:** Click on **Switch1** in your workspace, open the **CLI tab**, press Enter to wake up the prompt, and enter global configuration mode:

text

```
Switch1> enable
```

```
Switch1# configure terminal
```

- **Step 2: Enter the Connecting Interface:** Navigate to the specific physical interface that connects Switch1 directly to Switch2 (for example, Gigabit Ethernet port g0/1):

text

```
Switch1(config)# interface gigabitethernet 0/1
```

- **Step 3: Change the Interface Link Mode to Trunk:** Change the interface from a normal access mode link to a trunk link to allow multiple VLAN tags to pass simultaneously:

text

```
Switch1(config-if)# switchport mode trunk
```

- **Step 4: Explicitly Allow All VLAN Traffic:** Force the trunk port path to accept and carry the tag frames for all your newly deployed VLAN spaces:

text

```
Switch1(config-if)# switchport trunk allowed vlan all
```

```
Switch1(config-if)# end
```

```
Switch1# write memory
```

- **Step 5: Repeat Configurations on the Second Switch:** Open the CLI pane for **Switch2** and apply the identical trunk port setup commands on its connecting interface link:

text

```
Switch2> enable
```

```
Switch2# configure terminal
```

```
Switch2(config)# interface gigabitethernet 0/1
```

```
Switch2(config-if)# switchport mode trunk
```

```
Switch2(config-if)# switchport trunk allowed vlan all
```

```
Switch2(config-if)# end
```

```
Switch2# write memory
```

Question: 3

Add a router to your Packet Tracer network and configure Inter-VLAN routing so that a PC in VLAN 10 can successfully ping a PC in VLAN 20.

Hint: Use router-on-a-stick configuration with sub-interfaces for each VLAN.

Answer: 3

Steps to Configure Inter-VLAN Routing (Router-on-a-Stick)

To allow different VLANs (like VLAN 10 and VLAN 20) to communicate, you must route traffic through a layer-3 device. Follow these Cisco commands to set up sub-interfaces on a router interface (e.g., g0/0) linked to a trunk port on your main switch:

- **Step 1: Configure the Switch Trunk Link**

Connect your main switch interface (e.g., g0/2) to the router's interface (g0/0) using a copper straight-through cable. Configure that switch interface as a trunk port:

text

```
Switch> enable
```

```
Switch# configure terminal
```

```
Switch(config)# interface gigabitethernet 0/2
```

```
Switch(config-if)# switchport mode trunk
```

```
Switch(config-if)# exit
```

- **Step 2: Activate the Physical Router Interface**

Click on the Router device, open the **CLI tab**, press Enter, and turn on the main physical interface connection without giving it a primary IP:

text

```
Router> enable
```

```
Router# configure terminal
```

```
Router(config)# interface gigabitethernet 0/0
```

```
Router(config-if)# no shutdown
```

```
Router(config-if)# exit
```

- **Step 3: Create the Router Sub-Interface for VLAN 10**

Create the logic partition path, attach the 802.1Q encapsulation protocol tag, and assign the sub-interface default gateway IP address:

text

```
Router(config)# interface gigabitethernet 0/0.10
```

```
Router(config-subif)# encapsulation dot1Q 10
```

```
Router(config-subif)# ip address 192.168.1.1 255.255.255.0
```

```
Router(config-subif)# exit
```

- **Step 4: Create the Router Sub-Interface for VLAN 20**

Repeat the setup logic partition steps for your next active department zone block:

text

```
Router(config)# interface gigabitethernet 0/0.20
```

```
Router(config-subif)# encapsulation dot1Q 20
```

```
Router(config-subif)# ip address 192.168.2.1 255.255.255.0
```

```
Router(config-subif)# end
```

```
Router# write memory
```

- **Step 5: Set Default Gateways on the Client PCs**

Open the desktop network configuration panel on your workspace workstations and fill out their appropriate department gateway pathways:

- For **VLAN 10 PCs**, change the Default Gateway field to: **192.168.1.1**
- For **VLAN 20 PCs**, change the Default Gateway field to: **192.168.2.1**

Expected Result:

The initial one or two packets might drop due to the ARP discovery process, but you will immediately receive successful status readouts: **"Reply from 192.168.2.21: bytes=32 time<1ms TTL=127"**. This proves that inter-VLAN routing is fully active.

Question: 4

Set up static routes on the router so that a new network segment (e.g., a 'Creators' VLAN 40 on a third switch) can communicate with all existing VLANs, and test connectivity with ping.

Answer: 4

Steps to Configure Inter-VLAN Routing for VLAN 40

To add the new **"Creators" VLAN 40** to your router-on-a-stick network so it can communicate with all existing VLANs, follow these configuration steps:

- **Step 1: Define VLAN 40 on the Switches:** Open the CLI of your switches and add the new VLAN entry to the database:

text

```
Switch(config)# vlan 40
```

```
Switch(config-vlan)# name Creators
```

```
Switch(config-vlan)# exit
```

(Make sure the ports connecting the third switch to the rest of the network are configured as trunk ports).

- **Step 2: Create the Router Sub-Interface for VLAN 40:** Since the router interface is already physically connected to a switch trunk port and turned on, you only need to create the logical sub-interface path for the new segment:

text

```
Router> enable
```

```
Router# configure terminal
```

```
Router(config)# interface gigabitethernet 0/0.40
```

```
Router(config-subif)# description Default Gateway for Creators
```

```
Router(config-subif)# encapsulation dot1Q 40
```

```
Router(config-subif)# ip address 192.168.4.1 255.255.255.0
```

```
Router(config-subif)# end
```

```
Router# write memory
```

- **Step 3: Set the Default Gateway on Creators PCs:** Open the network configuration panel on the new VLAN 40 devices and set their gateway parameter:
 - **IP Address Range:** 192.168.4.10 to 192.168.4.30
 - **Subnet Mask:** 255.255.255.0
 - **Default Gateway:** 192.168.4.1

Verification Process

Because all sub-interfaces exist on the exact same physical router interface (g0/0), the router automatically builds **Connected Routes** for all segments (192.168.1.0/24, 192.168.2.0/24, and 192.168.4.0/24). No manual static route entries (ip route) are needed because the router handles the traffic natively.

To test global connectivity across the network segments:

1. Open the **Command Prompt** tool on a new Creators VLAN 40 PC (192.168.4.10).
2. Run a network ping test targeting an existing department PC (for example, a VLAN 10 PC):

```
cmd
```

```
ping 192.168.1.11
```

Expected Result:

The console screen will successfully output: **"Reply from 192.168.1.11: bytes=32 time<1ms TTL=127"**. This successful reply proves that the new network segment can communicate with all existing VLANs correctly.

Question: 5

Design a mini network in Packet Tracer that mimics a Zomato-style office: create VLANs for 'Support', 'Delivery', and 'Management', configure trunking and static routing, and document your topology with a screenshot and a brief explanation of how devices in different VLANs communicate.

Answer: 5

Zomato Office Network Topology Diagram

An illustrative layout of a multi-switch, segmented office workspace environment mimicking the requested corporate setup is shown below. You can save or reference this [Cisco Packet Tracer Network Topology](#) layout directly for your documentation:

How Devices in Different VLANs Communicate

Even though the endpoint devices are spread across multiple physical switches, traffic routing and isolation operate seamlessly through these specific network mechanics:

- **Layer 2 Isolation:** Devices within the same department (e.g., a Support PC on Switch 1 and a Support PC on Switch 2) share the same broadcast domain and can ping each other directly across the **Trunk Link** because their traffic frames carry identical VLAN tags. However, they cannot talk to any other department directly at Layer 2.

- **The Routing Boundary:** A "Support" PC cannot communicate directly with a "Delivery" or "Management" PC at the switch layer because switches block broadcasts crossing different VLAN borders. To bridge this gap, traffic must be sent up to a Layer 3 device.
- **Router-on-a-Stick Processing:** When a Support employee pings a Management server, the data packet is forwarded to its local default gateway interface on the **Core Router**. The router reads the incoming packet tag, references its internal **Connected Routing Table**, strips the old VLAN tag, appends the target destination's VLAN tag, and routes the data packet back down the trunk line to the target device.

